

Project 3 - Study Report

Topic: Proton VPN

Participant Description:

Demographics Table:  Usability Data Proton VPN

Participant Summary:

For this study, we were able to interview twelve participants through our four team members, with each of us finding three from our own fields. The ages ranged from as low as 21 to as high as 50, and the backgrounds were pretty varied. We had college students, a childcare center owner, and a restaurant manager. In terms of their tech comfort, the participants rated themselves between 4 and 9 on the 10-point scale, so we had a good mix of less and more experienced users. About half the users had used a VPN before, with services such as NordVPN, Surfshark, Opera VPN, and CISCO. In comparison, the majority of the participants had never used one before, providing a large range to efficiently analyze how Proton VPN would be for those with different proficiencies in VPN usage.

Study Design

Goal

The purpose of this study was to evaluate and see how intuitive Proton VPN is for everyday users, and also to understand how usability affects the way users perceive the app's security. Here's exactly what we wanted to see:

- To identify any usability issues ran into as they were navigating through the app's features, including connecting to a server, switching locations, and adjusting the settings.
- Observe how users were iterating with security features such as the kill switch, and whether or not they could find and understand them without guidance.
- Understand how participants are perceiving the security of their connection based on the app's visual feedback and design
- And lastly, capture any moments of confusion, hesitation, or frustration that could affect whether a user would trust or continue using the app

Measures

To evaluate usability, we collect quantitative and qualitative data from each session. The measures we tracked are:

- Task completion rate (yes/no)
- Time on task (seconds/minutes)
- Error and misclick count per task
- SEQ score after each task rated 1-7
- SUS score at the end of the session
- Observer notes on participant behavior and any think-aloud comments
- Participants perceived a sense of security

Tasks

All participants used a provided device with Proton VPN already installed and signed into a free account; nobody had to download anything or create an account. The task we gave the participants is:

1. Connect to a Server
2. Switch server location
3. Verify the VPN is working
4. Browse the web while connected (Things such as playing a video, visiting the website of the subject's choice, playing a free online game to test the connection)
5. Find and enable the kill switch
6. Check whether the VPN is active and secure

Interview Survey and Questions

After finishing all the tasks, we asked participants some post-study questions to get their comprehension of the app. These included things like whether any part felt confusing or frustrating to them, which task felt natural, whether the app made them feel secure, and whether they noticed any limitations. Participants also rated the app's ease of use on a scale of 1-7 and completed a full SUS survey. After each individual task, participants would rate how easy or difficult they felt on the SEQ scale.

How the Study was Conducted

Each session was run by a team member acting as the facilitator while also observing and taking notes of their participants. At the start of every session, the facilitator would read off a consent script so participants understood they were testing the app and not being evaluated themselves. The participants were also asked a few background questions about their age, tech comfort, and prior VPN experience before they started the task. Through the sessions, participants were asked to think out loud as they worked, saying what they were looking for and what they expected to happen. The session ran between 15 and 20 minutes per participant.

Study Results

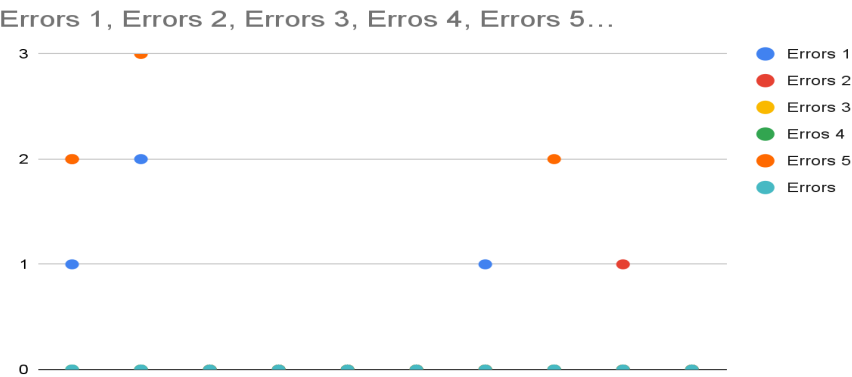
Task Completion Times

Most tasks were completed quickly overall. Task 1: Connecting to a server averaged about 13.5 seconds, and Task 2, switching server location, averaged around 15.4 seconds. Task 3: Verifying the VPN was working and checking if it was active and secure. Task 6 was the fastest, both averaging around 9 seconds. Browsing the web, Task 4 took the longest at an average of 48 seconds, which makes sense since participants were actually navigating the internet rather than just pressing a button. Finding and enabling the kill switch, Task 5 averaged 23.3 seconds and was noticeably slower than everything else, which lines up with what we saw in the usability issues.

Task	Average
Task 1	13.5 seconds
Task 2	15.4 seconds
Task 3	9.1 seconds
Task 4	52.8 seconds
Task 5	23.3 seconds
Task 6	11.1 seconds

Errors

The error rate for this study had a pretty low amount, Task 3,4,6 had zero errors through all the participants, Task 1 had 6 total errors, and Task 5 had 7 errors; both were coming from users who had low tech experience. The users either tapped on the wrong part or went into the wrong settings before they found what was needed.



SEQ Scores

The SEQ was high throughout all the tasks. Task 3 had the highest score at 6.82, meaning that all the participants found that verifying the VPN connection was the easiest task. Task 5 scored the lowest at 5.82, which lines up with the higher error counts and the longer completion times for that task. Tasks 1 and 2 were also slightly lower than the rest, which shows the learning curve that some of the participants had when first opening and using Proton.

Task	Average
Task 1	6.18
Task 2	6.27
Task 3	6.82
Task 4	6.50
Task 5	5.82
Task 6	6.09

Qualitative Observations

A few of the same themes that were appearing across all of our participants' sessions were that the kill switch was the most common issue for both low-tech comfort users and high-tech comfort users. The participants either couldn't find it or didn't understand what it did after they enabled it. The server switch task was also causing some issues, but it was mostly about not knowing which servers were free to use. But the best part is that the main connect button received good feedback for being obvious and easy to find. Most of the participants also thought that the green connection banner gave them some more reassurance that the app was working. Participants with prior VPN experience were able to move through the task more quickly and with fewer errors than those who had never used one.

Usability Issues

Kill Switch

Out of everything in the app, the kill switch gave participants the most trouble, and this came up across all four of our team members' sessions, not just one or two. Task 5 had the lowest average SE score at 5.82, the most total errors with 7, and the longest average completion time was 23.3 seconds. McKayla and Michelle both went to the wrong places before finding it, checking the account tab and profiles page, before they found the connection settings. McKayla said after the

session: “I couldn't find it and didn't know what it meant, even after I turned it on”. Michelle had about the same experience; she spent about a minute and a half making 3 errors along the way. But in Iris's session, her second participant had never seen a kill switch before and just thought it would cut their internet; they didn't realize it was meant to look like a light switch at all. Jazmin's participant found it fast but admitted they had skipped the explanation, so they still didn't know what it did, even though Drake's participants brought it up as something that needed to be fixed. The only person who had no trouble was Jalen, who already had experience with VPN's, and had gone straight to setting without any hesitation, which pretty much confirms that this feature only makes sense if you know what it is and know what you're looking for.

Server Switching and Free Tier

Another usability issue observed across participants was related to switching servers and understanding the limitations of the free version. While most users were able to eventually switch server locations, several initially tried interacting with the world map and expected it to be clickable, which confused when nothing happened. In addition, many participants were unsure which servers were actually available to them, as the app displays both free and paid options without clearly distinguishing between them. This led to frustration, especially when users attempted to select a server and were blocked by upgrade requirements. Even participants with prior VPN experience mentioned that the system felt unclear. Overall, this issue did not always prevent task completion, but it created unnecessary confusion and slowed users down, showing that the interface does not clearly communicate how server selection works in the free tier.

Security and Privacy Implications:

The usability problems identified in this study can have a direct impact on how well users are actually protected when using the VPN. One major issue was that some participants struggled to find or fully understand important features like the “kill switch.” This feature is designed to prevent data leaks if the VPN disconnects, but several users do not understand what it means or confuse it with simply turning off their internet connection. This misunderstanding means users could leave it off or misuse it and unknowingly expose their data. In addition, many users relied heavily on the lock icon and green “connected” banner to determine if they were secure. While this provides a sense of reassurance, it does not guarantee that users truly understand their level of protection, which creates a gap between perceived security and actual security.

Another issue observed in the interviews was that users were unsure whether the VPN was actually working in the background, even after connecting. Some participants reopened the app or considered checking online to confirm their connection. This uncertainty can lead users to either overtrust the app without a real understanding or feel unsure about their protection.

Additionally, limitations in the free version, such as restricted server access and unclear indicators of available servers, caused frustration. This could lead users to stop using the VPN or not use it consistently. Overall, even if the app is technically secure, these usability issues make it harder for users to use it correctly, which reduces the effectiveness of its security and privacy protections.

Recommendations: Suggestions based on usability issues

Increase Accessibility and Learnability for the Kill Switch Tool

A frequent concern among the participants was the lack of easy access to the kill switch and the overall confusion in utilizing the tool. A participant stated that they had never learned about the kill switch, even though they had used a VPN before. The best fix for this would be adding a short prompt the first time someone connects, something that points them to the kill switch and will explain why it matters in a more plain tone. A participant even suggested that turning it on by default ensures users are protected whether they find it or not. Another suggestion would be to make it more noticeable among the different buttons available, especially given how important it is for users to stay protected when browsing.

Ensuring Users are more Knowledgeable of Available Free Options

Another suggestion would be to clarify which options are available for free users, especially with the World Map and its available server options. When choosing and changing server options, participants felt confused about what could be accessed, as it seemed most options were available. Even participants with previous VPN knowledge felt a bit confused about the system's availability. To combat this issue, users should be presented with the available options being highlighted on the world map. Another suggestion would be to give a prompt on what servers can be accessed when pressing the "Change Server" button on the left side. These refinements should lessen the frustrations regarding the application's limitations in free content.

Overall Walkthrough and Proper Tutorial

While Proton VPN provided the basics of a VPN application, it failed to provide a proper walkthrough for new users. The lack of a tutorial confused first-time users, especially since they made up more than half of the study's participants. A participant also specifically stated that there should be a walkthrough on the options and better labeling for tools like the kill switch. One option could be to ask users at the beginning whether they have used a VPN before and whether they would like to complete an application tutorial. These improvements can help make the application clearer for both experienced and inexperienced users.

Clarify how to set up free accounts.

While not part of the main study, some participants had to set up the vpn themselves due to doing the study over the internet. A common theme among those participants is that when setting up a free account. It was hard to determine if it was even possible to set up a free account in the first place. This led to massive confusion in setting up the application to begin with. A massive improvement would be to clarify how to set up a free account. Being able to access the free account without confusion.